ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)

path to success

JEE (MAIN+ADVANCED)

Organic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Chemistry in Everyday Life

English Medium

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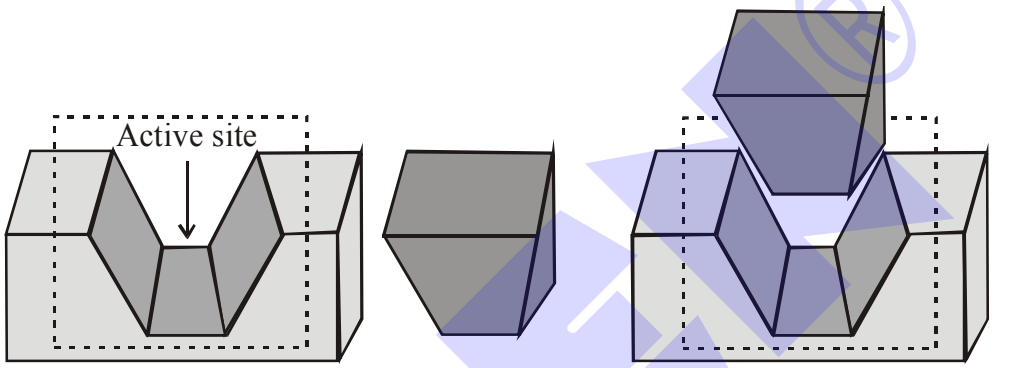
CHEMISTRY IN EVERYDAY LIFE

Drug-Target Interaction

Macromolecules of biological origin perform various functions in the body. For example, proteins which perform the role of biological catalysts in the body are called **enzymes**, so enzymes are biocatalyst and they are generally proteinaceous in nature.

Enzymes as Drug Targets

(a) Catalytic action of enzymes



(a) Enzyme (b) Substrate (c) Enzyme holding substrate

In their catalytic activity, enzymes perform two major functions:

(i) The first function of an enzyme is to hold the substrate for a chemical reaction. Active sites of enzymes hold the substrate molecule in a suitable position, so that it can be attacked by the reagent effectively.

Substrates bind to the active site of the enzyme through a variety of interactions such as **Ionic bonding, Hydrogen bonding,** Van-der Waalsinteraction or Dipole-interaction.

(ii) The second function of an enzyme is to provide functional groups that will attack the substrate and carry out chemical reaction.

(b) Drug-enzyme interaction

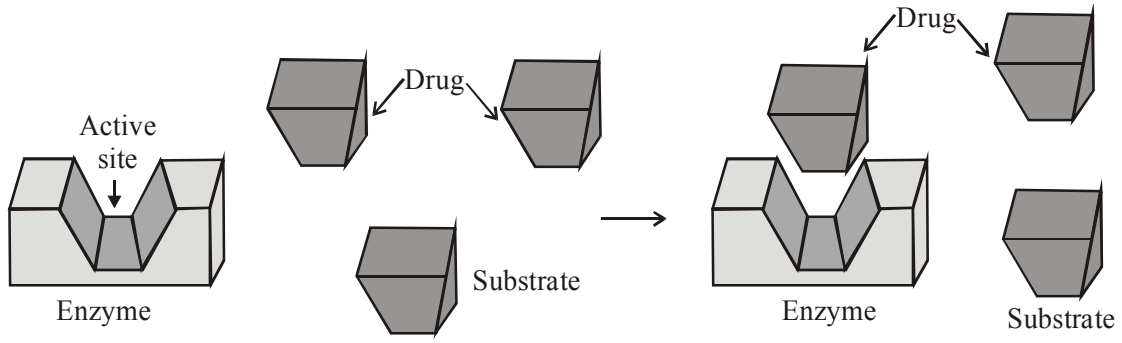
Drugs can block the binding site of the enzyme and prevent the binding of substrate.

Drugs can inhibit the catalytic activity of the enzyme. Such drugs are called **enzyme inhibitors.**

Drugs inhibit the attachment of substrate on active site of enzymes in two different ways;

(i) Drugs compete with the natural substrate for their attachment on the active sites of enzymes. Such drugs are called **competitive inhibitors**

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Drug and substrate competing for active site of enzyme

Drug blocks the active site of enzyme (Drug is competitive inhibitor which is similar in shape and size of substrate)

(ii) Some drugs do not bind to the enzyme's active site. These bind to a different site of enzyme which is called **allosteric site.**

This binding of inhibitor at allosteric site changes the shape of the active site in such a way that substrate cannot recognise it.

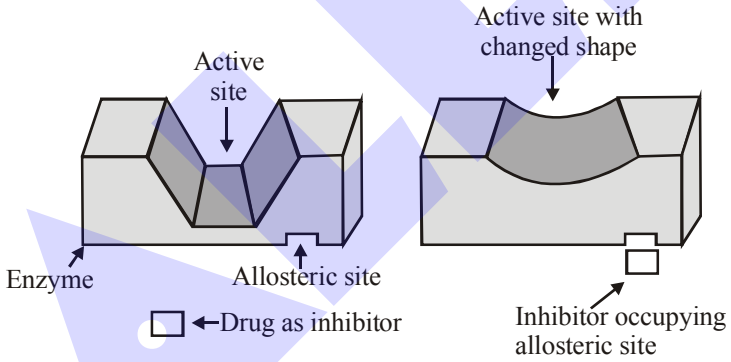


Fig. : *Non-competitive inhibitor changes the active site of enzyme after binding at allosteric site.*

- The body then degrades the enzyme-inhibitor complex and synthesises the new enzyme

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PART A : ANALGESIC DRUGS

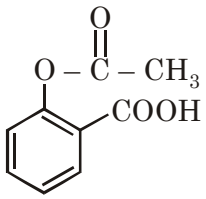
ANALGESIC DRUGS

(a) Non Narcotic Analgesic

(b) Narcotic Analgesic

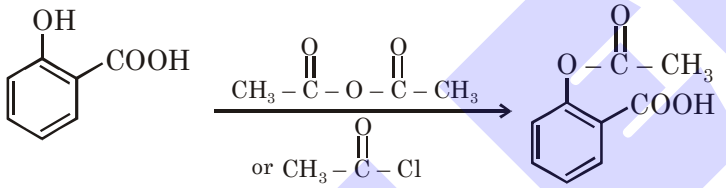
(a) NON NARCOTIC ANALGESIC

1. ASPIRIN (IUPAC Name : 2-Ethanoyloxybenzoic acid)



I. Medical use : Analgesic (Non narcotic / Non addictive) & Antipyretic.

II. Prepration : Aspirin is prepared by acetylation of salicylic acid



Salicylic acid (SNAE) Aspirin

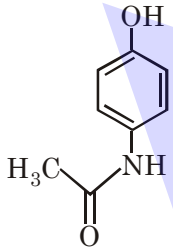
III. Functional group : Acid and ester

IV. Test of Functional group : All +ve tests for carboxylic acids

V. Aromaticity : Aromatic (Homocyclic)

VI. DOU : 6

2. PARACETAMOL (IUPAC Name : N-(4-Hydroxyphenyl)ethanamide



I. Medical use : Analgesic (Non-narcotic / non-addictive) & Antipyretic.

II. Hybridisation state :

$sp^2C \rightarrow 7C$

$sp^3C \rightarrow 1C$

III. Functional group : Phenolic OH, secondary amide

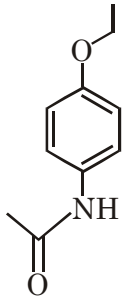
IV. Test of Functional group : +ve test with neutral $FeCl_3$

V. Aromaticity : Aromatic (Homocyclic)

VI. DOU : 5

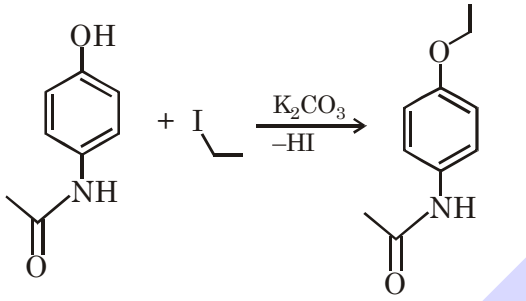
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3. PHENACITIN (IUPAC Name : N-(4-Ethoxyphenyl)ethanamide



I. Medical use : Analgesic (Non-narcotic / non-addictive) & Antipyretic.

II. Prepration :



Paracetamol Phenacitin

III. Hybridisation state

$sp^2C \rightarrow 7C$

$sp^3C \rightarrow 3C$

IV. Functional group / test : Ether and 2° amide / –ve test with neutral $FeCl_3$

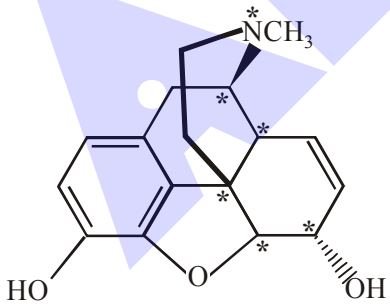
V. Aromaticity : Aromatic (Homocyclic)

VI. DOU : 5

NOTE : Quinine, Chloroquine, Paraquine and Primaquine are used as antimalaria.

(b) NARCOTIC ANALGESIC

1. MORPHINE



I. Medical use : Morphine is used for reliefe a post-operative pain, cardiac pan, child birth and pains of terminal cancer

II. Number of chiral centre : 6

III. Number of chiral carbon : 5

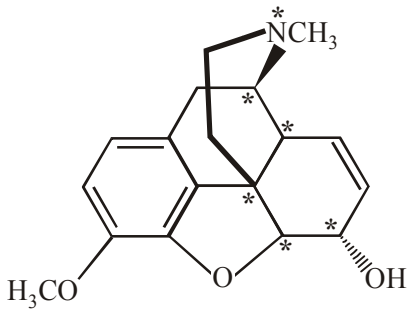
IV. Functional group : Morphin narcotics are also called opiates.

V. Aromaticity : Aromatic (Heterocyclic)

VI. DOU : 9

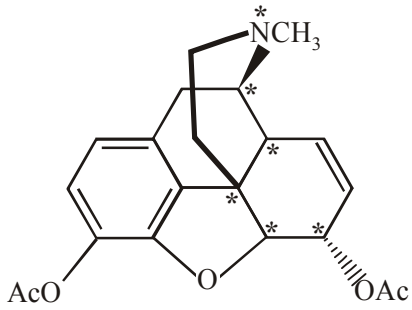
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2. CODEINE



I. DOU : 9

3. HEROIN



I. DOU : 11

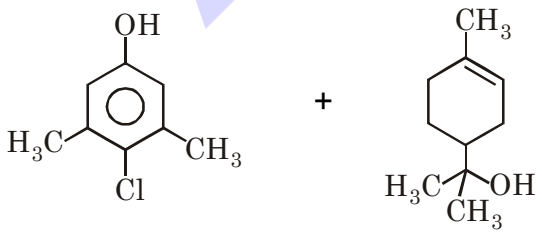
	Phenolic OH	Alcohol
Morphine	+	+
Codeine	–	+
Heroin	–	–

Acidic strength order : Morphine > Codeine > Heroin

PART B : ANTISEPTIC, DISINFECTANTS & ANTI-FERTILITY DRUG

1. ANTISEPTIC

I. Dettol (Chloroxylenol + Terpineol)

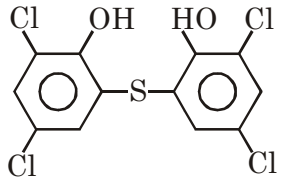


Chloroxylenol **Terpineol**

Medical use : Dettol, Soframicin are used as anticeptic

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II. Bithionol

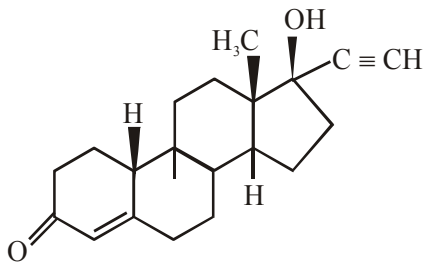


Medical use : Bithionol is added to soaps for anticeptic properties.

2. NOTE : 1% solution of phenol in disinfectant while 0.2% solution of phenol is antiseptic.

3. ANTI-FERTILITY DRUGS

I



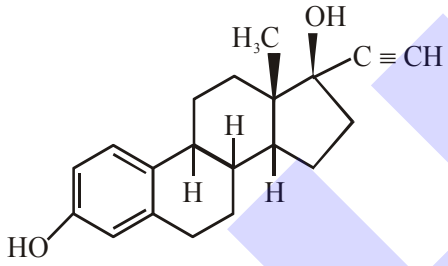
Norethindorne

Medical Use : It is present in birth control pils for family planning.

Number of Chiral carbons : 6

Functional groups : Ketone, Alcoholic-OH and terminal alkyine

II



Ethynylestradiol (novestrol)

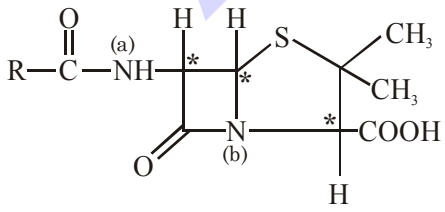
Medical Use : It is present in birth control pils for family planning.

Number of Chiral carbons : 5

Functional groups : Phenolic-OH, Alcoholic-OH and terminal alkyine

PART C : ANTIBIOTICS

1. PENICILLIN



I. Medical use : Bactericidal Antibacterial (Killing effect on bacteria)

NOTE : Ampicllin and Amoxycillin are synthetic modifications of Penicilline

II. Number of chiral carbon centre : 3

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III. Functional group :

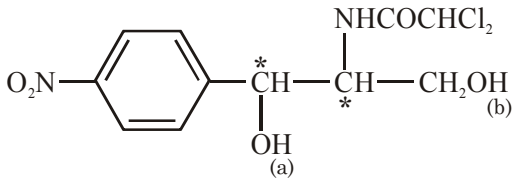
(i) Carboxylic Acid (ii) 2°Amide (iii) 3° Amide (iv) Thioether

IV. Aromaticity : Non-aromatic (Heterocyclic)

V. DOU : 5

VI. Basicity : b > a

2. CHLORAMPHENICOL



I. Medical use :

(i) Bacteriostatic antibiotic (Static / inhibitory effect on microbes)

(ii) Broad spectrum antibiotic (Covers wide range of diseases like typhoid, dysentery, acute fever, certain form of urinary infections, meningitis and pneumonia)

II. Number of chiral carbon atoms : 2

III. TSI : 4 (OA)

IV. Functional group :

(i) Alcohol (ii) Nitro (iii) 2° amide (iv) Chloro

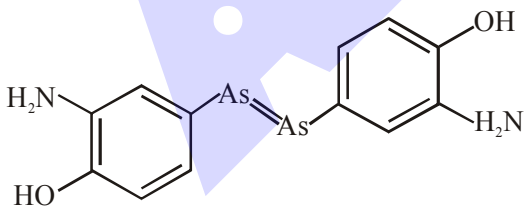
V. Aromaticity : Aromatic (Heterocyclic)

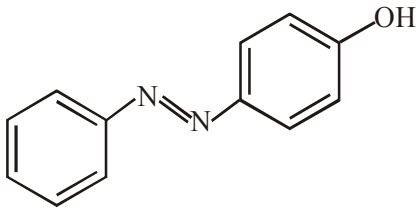
VI. DOU : 6

VII. Acidic strength : a > b

NOTE : DYSIDAZIRINE antibiotic is toxic towards certain strains of cancer cells.

3. SALVARSAN **Structure similar to azodye**





I. Medical use :

(i) Effective for treatment against spirocheti (bacteria that causes syphilis)

(ii) Toxic to human beings

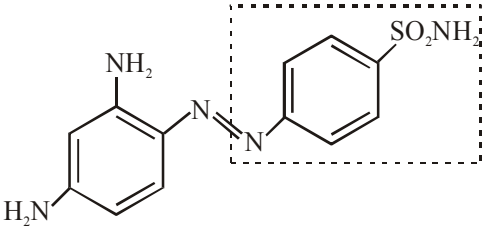
II. Test of Functional group : Phenolic OH (+ve test with neutral $FeCl_3$), azodye test

III. Aromaticity : Aromatic (Homocyclic)

IV. DOU : 9

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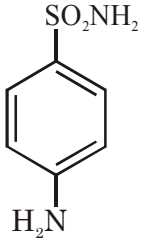
4. PRONTOSIL



I. Medical use :

(i) First effective antibacterial agent

(ii) Converted to sulphanilamide in body which is the real active compound

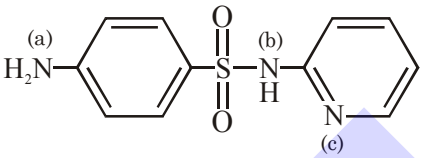


(Sulphanilamide)

II. Aromaticity : Aromatic (Homocyclic)

III. DOU : 11

5. SULPHAPYRIDINE



I. Medical use : Sulphadrug

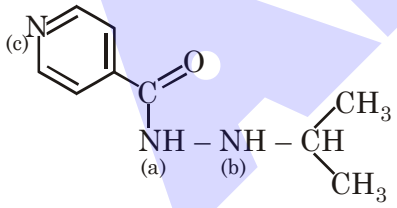
II. Aromaticity : Aromatic (Heterocyclic)

III. DOU : 10

IV. Most Basic site : N_C

PART D : ANTIDEPRESSANT & TRANSQUILIZERS

1. IPRONIAZID



I. Medical use :

(i) Mood elevators (Antidepressant drugs)

(ii) Catalyse the degradation of noradrenaline

II. Number of chiral centre : 1

III. Hybridisation state

$sp^2C \rightarrow 6C$

$sp^3C \rightarrow 3C$

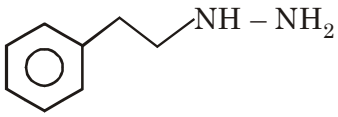
IV. Aromaticity : Aromatic (Heterocyclic)

V. DOU : 5

VI. Basicity : b > c > a

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2. PHENELZINE (Nardil)



I. Medical use :

(i) Mood elevators (Antidepressants)

(ii) Catalyse the degradation of noradrenaline

II. Number of chiral centre : 1

III. Hybridisation state

$sp^2C \rightarrow 6$

$sp^3C \rightarrow 2$

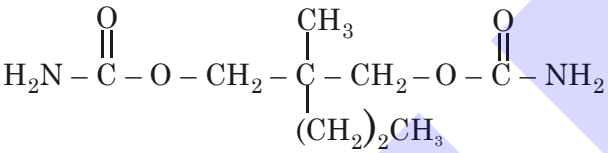
IV. Functional group : Amine

V. Test of Functional group : Diazotization (NH_2)

VI. Aromaticity : Aromatic (Homocyclic)

VII. DOU : 4

3. MEPROBAMATE



I. Medical use : Mild tranquilizer

II. Number of chiral centre : 0

III. Hybridisation state :

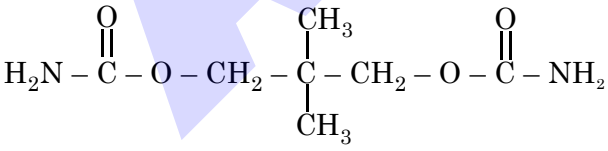
$sp^2C \rightarrow 2$

$sp^3C \rightarrow 7$

IV. Aromaticity : Non aromatic

V. DOU : 2

4. EQUANIL



I. Medical use : Controlling depression and hypertension

II. Number of chiral centre : 0

III. Hybridisation state :

$sp^2C \rightarrow 2$

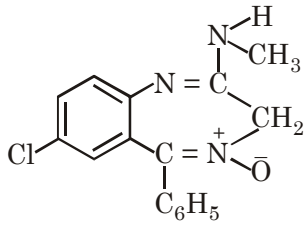
$sp^3C \rightarrow 5$

IV. Aromaticity : Non aromatic

V. DOU : 2

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5. CHLORDIAZEPOXIDE



I. Medical use : Mild tranquilizer

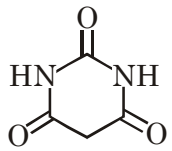
II. Hybridisation state

$sp^2C \rightarrow 14$ $sp^3C \rightarrow 2$

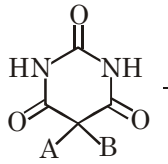
III. Aromaticity : Aromatic (Heterocyclic)

IV. DOU : 11

6. BARBITURIC ACID



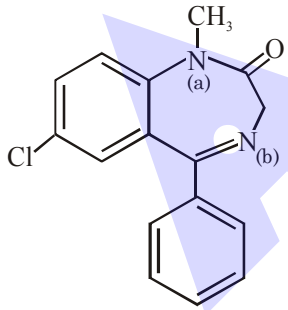
7. BARBITURATES



		A	B
1	Veronal	Ethyl	Ethyl
2	Luminal	Ethyl	Phenyl
3	Amytal	Ethyl	Isopentyl
4	Nembutal	Ethyl	Secondary active amyl
5	Seconal	Allyl	Secondary active amyl

They are derivatives of barbituric acid which are used as hypnotic (Sleep producing agents)
Tautomerism : Shows Tautomerism

8. VALIUM



I. Medical use : Tranquilizer

II. Hybridisation state :

$sp^2C \rightarrow 14$

$sp^3C \rightarrow 2$

III. Test of Functional group :

–ve test with $AgNO_3$ (aq.)

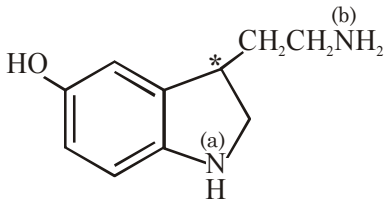
IV. Aromaticity : Aromatic (Heterocyclic)

V. DOU : 11

VI. Basicity : b > a

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9. SEROTONIN



I. Medical use : Tranquilizer

II. Number of chiral centre : 1

III. TSI : 2 (OA)

IV. Hybridisation state :

$sp^2C \rightarrow 6$

$sp^3C \rightarrow$ **4**

V. Functional group :

1° Amine

2° Amine

Phenolic OH

VI. Test of Functional group :

+ve isocyanide test

+ve test with neutral $FeCl_3$

Yellow dye with benzene diazonium chloride

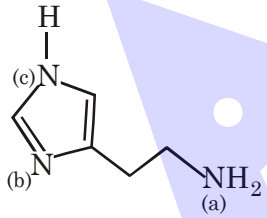
VII. Aromaticity : Aromatic (Heterocyclic)

VIII. DOU : 5

IX. Basicity : b > a

PART E : HISTAMINE, ANTI-HISTAMINE & ANTACIDS

(a) HISTAMINE



I. Medical use:

(i) Stimulates secretion of pepsin and HCl in stomach

(ii) Contracts the muscles in bronchi and gut

(iii) Relaxes blood vesscls (Vasodilator)

(iv) Nasal congestion and allergic reaction to pollen

II. Number of chiral centre : 0

III. Hybridisation state:

$sp^2C \rightarrow 3C$

$sp^3C \rightarrow 2C$

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IV. Functional group : Amine

V. Test of Functional group : Isocyanide test

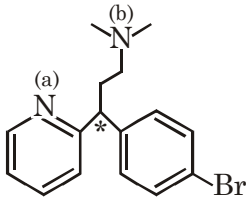
VI. Aromaticity : Aromatic (Heterocyclic)

VII. DOU : 3

VIII. Basicity b > a > c

(b) ANTI HISTAMINE

1. BROMPHENIRAMINE (Dimetapp, Dimetane)



I. Medical use : Antihistamine

II. Number of chiral centre : 1

III. TSI : 2 (OA)

IV. Hybridisation state :

$sp^2C \rightarrow 11C$

$sp^3C \rightarrow 5C$

V. Functional group : 3° amine, aryl bromide

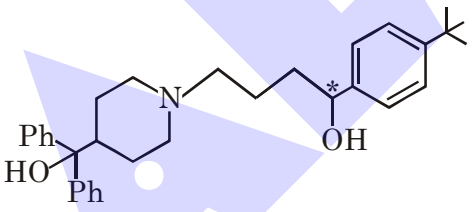
VI. Test of Functional group : –ve test with $AgNO_3$ (aq.)

VII. Aromaticity : Aromatic (Heterocyclic)

VIII. DOU : 8

IX. Basicity : $N_b > N_a$

2. TERFENADINE (Seldane)



I. Medical use : Antihistamine

II. Number of chiral centre : 1

III. TSI : 2

IV. Hybridisation state :

$sp^2C \rightarrow 18\ sp^3C \rightarrow 14$

V. Functional group : 3° amine, alcohol

VI. Test of Functional group :

(i) Positive test of alcoholic OH (Including Lucas test)

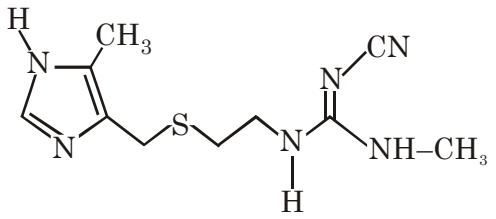
VII. Aromaticity : Aromatic (Heterocyclic)

VIII. DOU : 13

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(c) ANTACIDS

1. CIMETIDINE (Tegamet)



I. Medical use :

(i) Antacid (against acidity)

II. Number of chiral centre : 0

III. TSI : 2 (GI)

IV. Hybridisation state :

spC $\rightarrow$ 1C

$sp^2C \rightarrow$ 4C

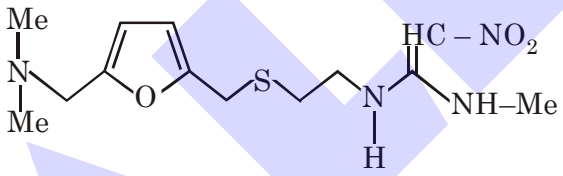
$sp^3C \rightarrow$ 5C

V. Functional group : Thioether, Cyanide etc.

VI. Aromaticity : Aromatic (Heterocyclic)

VII. DOU : 6

2. RANITIDINE (Zantac)



I. Medical use : Antacid

II. Number of chiral centre : 0

III. TSI : 2 (GI)

IV. Hybridisation state :

$sp^2C \rightarrow$ 6C

$sp^3C \rightarrow$ 7C

V. Functional group : Amines, Thiol, nitro etc.

VI. DOU : 5

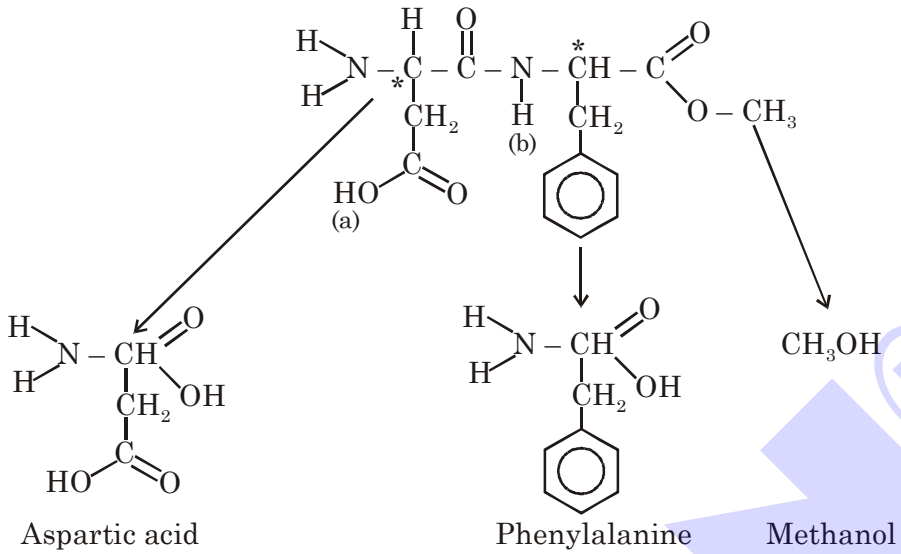
VII. Aromaticity : Aromatic (Heterocyclic)

NOTE : $Al(OH)_3$, $Mg(OH)_2$ are also used as antacids

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PART F : ARTIFICIAL SWEETNERS

1. ASPARTAME



I. Use :

(i) Used as an artificial sweetener (100 times sweeter than cane sugar)

(ii) Methyl ester of dipeptide formed from aspartic acid and phenylalanine

(iii) Use of aspartame is limited to cold food and soft drinks because it is unstable at cooking temperature

II. Number of chiral centre: 2

III. TSI : 4

IV. Hybridisation state :

$sp^2C \rightarrow 9$

$sp^3C \rightarrow 5$

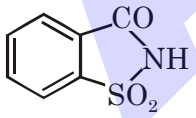
V. Functional group : Acid, Ester, Amide

VI. Aromaticity : Aromatic (Homocyclic)

VII. DOU : 7

VIII. Acidic strength : a > b

2. SACCHARIN (Ortho-sulphobenzimide)



I. Use :

(i) Used as an artificial sweetener (550 times sweeter than cane sugar)

(ii) Excreted from the body in urine unchanged.

II. Number of chiral centre : 0

III. Hybridisation state :

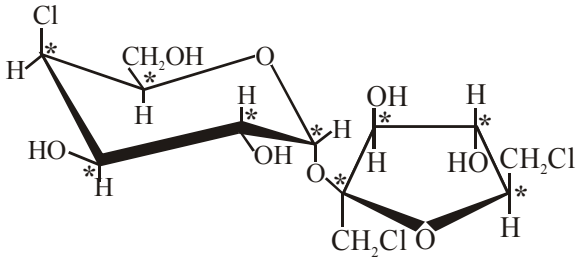
$sp^2C \rightarrow 7$

IV. Aromaticity : Aromatic (Heterocyclic)

V. DOU : 8

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3. SUCRALOSE



I. Use :

(i) Used as an artificial sweetener (600 times sweeter than cane sugar)

(ii) Appearance and taste like sugar

(iii) Stable at cooking temperature

(iv) Does not provide calories

(v) Trichloro derivative of sucrose

II. Number of chiral centre : 9

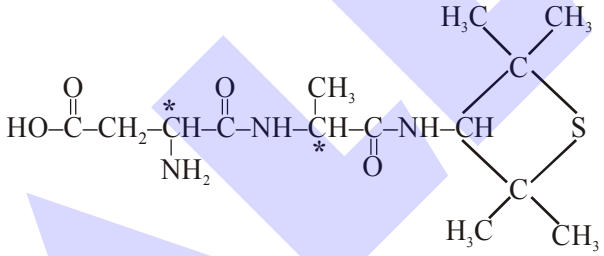
III. Functional group : Alcohol, ether, alkyl halide

IV. Aromaticity : Non aromatic (Heterocyclic)

V. DOU : 2

VI. Hybridisation : $sp^3C \rightarrow 12C$

4. ALITAME



I. Use :

(i) High potency sweetener (2000 times sweeter than cane sugar)

(ii) Although it is more stable than aspartame, the control of sweetness of food is difficult while using it.

II. Number of chiral centre : 2

III. Hybridisation state :

$sp^2C \rightarrow 3$

$sp^3C \rightarrow 11$

IV. Functional group : Acid, amide, amine

V. Aromaticity : Non-aromatic (Heterocyclic)

VI. DOU : 4

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PART G : SOAP, DETERGENT & PRESERVATIVES

(a) SOAPS

Sodium salt of

1. Stearic acid

$nC_{17}H_{35}CO_2H$

2. Oleic acid

$nC_{17}H_{33}CO_2H$

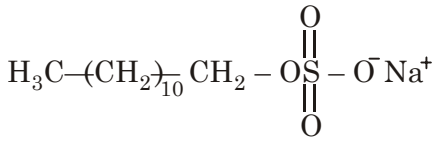
3. Palmitic acid

$nC_{15}H_{31}CO_2H$

(b) SYNTHETIC DETERGENTS

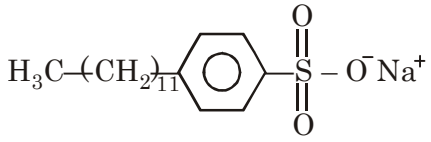
1. Anionic

i.



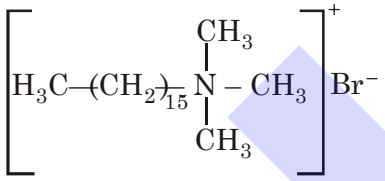
Sodium laurylsulphate

ii.



Sodium dodecylbenzenesulphonate

2. Cationic



Cetyltrimethylammonium bromide

i Non ionic

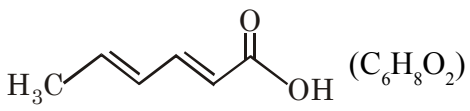
$CH_3(CH_2)_{16}COO(CH_2CH_2O)_nCH_2CH_2OH$

(non-ionic detergent), it is dishwashing detergent

Note : Synthetic detergents are better than soaps because synthesis detergents gives foam in both soft and hard water while soap do not give foam in hard water.

(c) FOOD PRESERVATIVES

1. SORBIC ACID



I. Use :

Salts of sorbic acid and propanoic acid are used as food preservatives (prevent microbial growth on food)

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II. Number of chiral centre : 0

III. TSI : 4 (GI)

IV. Hybridisation state :

$sp^2C \rightarrow 5$

$sp^3C \rightarrow 1$

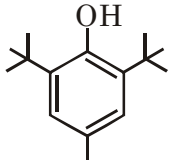
V. Functional group : Carboxylic acid

VI. Aromaticity : Non-aromatic

VII. DOU : 3

2. Butylatedhydroxytoluene (BHT)

(IUPAC Name : 2,6-Bis(1,1-dimethylethyl)-4-methylphenol)



I. Use :

(i) Used as antioxidants (Help in food preservation by retarding the action of oxygen on food)

(ii) Sometimes citric acid added along BHT to produce more effect

II. Number of chiral centre : 0

III. Hybridisation state :

$sp^2C \rightarrow 6$

$sp^3C \rightarrow 9$

IV. Functional group : Phenolic OH (SIR)

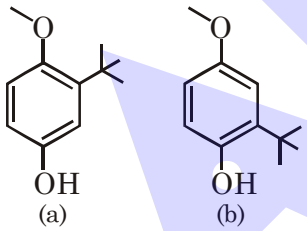
V. Aromaticity : Aromatic (Homocyclic)

VI. DOU : 4

3. Butylated hydroxy anisole (BHA)

(IUPAC Name : (a) 3-(1,1-Dimethylethyl)-4-methoxyphenol

(b) 2-(1,1-Dimethylethyl)-4-methoxyphenol)



(a) (b)

I. Use :

(i) Used as antioxidants (Help in food preservation by retarding the action of oxygen on food)

(ii) Sometimes citric acid added along BHA to produce more effect

II. Number of chiral centre : 0

III. Hybridisation state :

$sp^2C \rightarrow 6$

$sp^3C \rightarrow 5$

IV. Functional group : Phenolic OH, ether

V. Aromaticity : Aromatic (Homocyclic)

VI. DOU : 4

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